

BING-YUE WU

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EDUCATION

Arizona State University	2023 – Present
<i>Ph.D. in Electrical Engineering</i>	<i>Tempe, Arizona</i>
National Taiwan University of Science and Technology (Taiwan Tech, NTUST)	2021 – 2023
<i>M.S. in Electrical Engineering</i>	<i>Taipei, Taiwan</i>
National Taiwan University of Science and Technology (Taiwan Tech, NTUST)	2016 – 2020
<i>B.S. with major in Electrical Engineering and minor in Computer Science and Information Engineering</i>	<i>Taipei, Taiwan</i>

WORK EXPERIENCE

Arizona State University	Aug. 2023 – Present
<i>Graduate Research Assistant</i>	<i>Tempe, Arizona</i>
- Developed a large-scale LLM dataset for physical design tool (OpenROAD) interaction, built a domain-specific LLM to generate OpenROAD scripts and answer OpenROAD-related queries, and created a self-correcting agent that iteratively refines scripts using real tool feedback. - Proposed a diffusion-based generative framework to generate synthetic circuit layout heatmaps and open-sourced 23K+ synthetic samples to address data scarcity in ML-driven physical design research. - Investigated LLM-based automated design rule violation (DRV) fixing and developed benchmarking infrastructure to evaluate LLM-driven DRC methodologies. - Served as Topic Chair for ICCAD 2024 CAD Contest (Problem C) and contributed to OpenROAD/CircuitOps infrastructure to advance ML-EDA research and education.	
Synopsys Inc.	Oct. 2021 – June 2022
<i>Intern (Technical-Engineering)</i>	<i>Taipei, Taiwan</i>
- Researched a novel Machine Learning-based (ML) solution estimating the effective resistance of Power Delivery Networks in advanced VLSI designs to speed up runtime and improve the accuracy of ML-driven IR analysis tools. - Responsible for designing the data pipeline, the ML model architecture, and the entire effective resistance estimation workflow.	
Academia Sinica	Jul. 2020 – Nov. 2020
<i>Full-time Research Assistant at Computational Finance and Data Analytics Lab</i>	<i>Taipei, Taiwan</i>
- Conducted research on a novel Transformer Encoder-based model with financial number category awareness, designing new pre-training and fine-tuning tasks to enhance its performance. - Developed an Online Loan Application Recommender System for E.SUN Commercial Bank, achieving nearly 300% performance improvement. Responsibilities included workflow design, model development, and building Python APIs for industrial deployment.	

SKILLS

Programming Languages: C, C++, Python, Tcl
EDA Tools: OpenROAD, OpenSTA, Innovus, Genus, ICC2, Design Compiler, HSpice, Virtuoso, Calibre
Language Ability: Mandarin (Native), English (Fluent)

PUBLICATIONS

- B.-Y. Wu, C.-T. Ho, H. Yang, C.-C. Chang, A. B. Akkur, B. Khailany, and V. A. Chhabria, “Special Research Session: Toward Agentic Solution for DRC Challenges in Digital VLSI Design”, in *Proc. DAC*, 2026. [[GitHub](#)]**
- B.-Y. Wu, A. Dey, A. Rovinski, and V. A. Chhabria, “Focus Session: Large Language Models in Physical Design: From Data Generation to Intelligent Agents”, in *Proc. DATE*, 2026.**
- B.-Y. Wu and V. A. Chhabria, “DALI-PD: Diffusion-based Synthetic Layout Heatmap Generation for ML in Physical Design”, in *Proc. ASP-DAC*, 2026. [[GitHub](#)]**
- V. A. Chhabria, A. Ghose, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Y. Liu, Z. Wang, and B.-Y. Wu, “IEEE DATC RDF-2025: Enabling an EDA Research Ecosystem”, in *Proc. ICCAD*, 2025.**

- B.-Y. Wu**, U. Sharma, A. Rovinski, and V. A. Chhabria, “**OpenROAD Agent: An Intelligent Self-Correcting Script Generator for OpenROAD**”, in *Proc. ICLAD*, 2025. [[GitHub](#)]
- V. A. Chhabria, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Z. Wang, **B.-Y. Wu**, and D. Yoon, “**Strengthening the Foundations of IC Physical Design and ML EDA Research**”, in *Proc. ICCAD*, 2024.
- V. A. Chhabria, **B.-Y. Wu**, U. Sharma, K. Kunal, A. Rovinski, and S. S. Sapatnekar, “**Generative Methods in EDA: Innovations in Dataset Generation and EDA Tool Assistants**”, in *Proc. ICCAD*, 2024.
- B.-Y. Wu**, R. Liang, G. Pradipta, A. Agnesina, H. Ren, and V. A. Chhabria, “**2024 ICCAD CAD Contest Problem C: Scalable Logic Gate Sizing Using ML Techniques and GPU Acceleration**”, in *Proc. ICCAD*, 2024. [[GitHub](#)]
- U. Sharma*, **B.-Y. Wu***, S. R. D. Kankipati, V. A. Chhabria, and A. Rovinski, “**OpenROAD-Assistant: An Open-Source Large Language Model for Physical Design Tasks**”, in *Proc. MLCAD*, 2024. [[GitHub](#)]
- V. Gopalakrishnan, **B.-Y. Wu**, and V. A. Chhabria, “**ML-INSIGHT: Machine Learning for Inrush Current Prediction and Power Switch Network Improvement**”, in *Proc. ISLPED*, 2024.
- B.-Y. Wu**, U. Sharma, S. R. D. Kankipati, A. Yadav, B. K. George, S. R. Guntupalli, A. Rovinski, and V. A. Chhabria, “**EDA Corpus: A Large Language Model Dataset for Enhanced Interaction with OpenROAD**”, in *Proc. LAD*, 2024. (Best Paper Nominated) [[GitHub](#)]
- V. A. Chhabria, W. Jiang, A. B. Kahng, R. Liang, H. Ren, S. S. Sapatnekar, and **B.-Y. Wu***, “**OpenROAD and CircuitOps: Infrastructure for ML EDA Research and Education**”, in *Proc. VTS*, 2024. (Primary Author) [[GitHub](#)]
- B.-Y. Wu**, S.-Y. Fang, H.-W. Chang, and P. Wei, “**SpeedER: A Supervised Encoder-Decoder Driven Engine for Effective Resistance Estimation of Power Delivery Networks**”, in *Proc. MLCAD*, 2022. (Best Paper Award)

PROJECT EXPERIENCE

- Slicing Floorplanner with Simulated Annealing** | Python May 2024
- Built a simulated annealing engine from scratch to generate area-minimized slicing floorplans for hard and soft macro blocks, supporting up to 500 blocks.
 - Modeled floorplans as Polish expressions and designed move operators (operand swap, cut complement, adjacent operand operator swap) to guarantee valid state transitions and complete exploration of the solution space.
- Static Timing Analysis Engine** | Python Mar. 2024
- Parsed ISCAS85 circuit netlists and liberty files to build directed graph representations, and performed NLDM lookup-table interpolation for gate delay and output slew computation.
 - Implemented topological-order forward and backward traversals to compute arrival times, required arrival times, slack values, and critical paths for circuits with up to 100K gates.
- ASIC Acceleration for Graph Convolutional Networks** | Verilog, Design Compiler, Innovus Dec. 2023
- Designed a GCN accelerator in Verilog targeting sub-100ns end-to-end latency, and verified through RTL simulation.
 - Performed logic synthesis, place-and-route, and conducted post-routing simulation and power measurement to validate timing and power targets.
- Standard Cell Design in 7nm ASAP7 PDK** | Virtuoso, Calibre, HSpice Oct. 2023
- Designed standard cell schematic and layout using the ASAP7 7nm FinFET PDK in Cadence Virtuoso, adhering to the 7.5-track cell architecture and design rule constraints.
 - Verified layout with Calibre DRC/LVS, performed parasitic extraction, and compared pre- and post-layout HSpice simulation results under FO4 loading conditions.

AWARDS

- MLCAD Student Travel Grant** Sept. 2024
- Ferdinand A. Stanchi Fellowship** Aug. 2024
- DAC Young Fellow Travel Grant** Jun. 2024
- MLCAD Student Travel Grant** Sept. 2023
- Fulton Fellows Fellowship** Aug. 2023
- Best Paper Award at MLCAD 2022** Sept. 2022

PROFESSIONAL EXPERIENCE

- Topic Chair of Problem C at 2024 ICCAD CAD Contest** Oct. 2024
- IEEE CEDA* Newark, New Jersey
- 2024 ASP-DAC Tutorial Talk** Jan. 2024
- ACM SIGDA* Incheon, South Korea